VortexTech Cybersecurity Internship

Week 2 Report

Hands-on with Basic Security Tools

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Track: Cybersecurity Internship
Week: 2

1. Objective

The objective of this task was to gain practical experience with two fundamental cybersecurity concepts: password security and network reconnaissance. A Python-based password strength checker was used to evaluate password complexity, while Nmap was used to scan localhost for open ports. This exercise helped demonstrate how weak passwords and exposed network services can increase security risks.

2. Password Strength Checker

Overview

A Python program was used to evaluate password strength based on multiple security criteria. The program analyzes user-entered passwords and provides a strength rating along with suggestions for improvement.

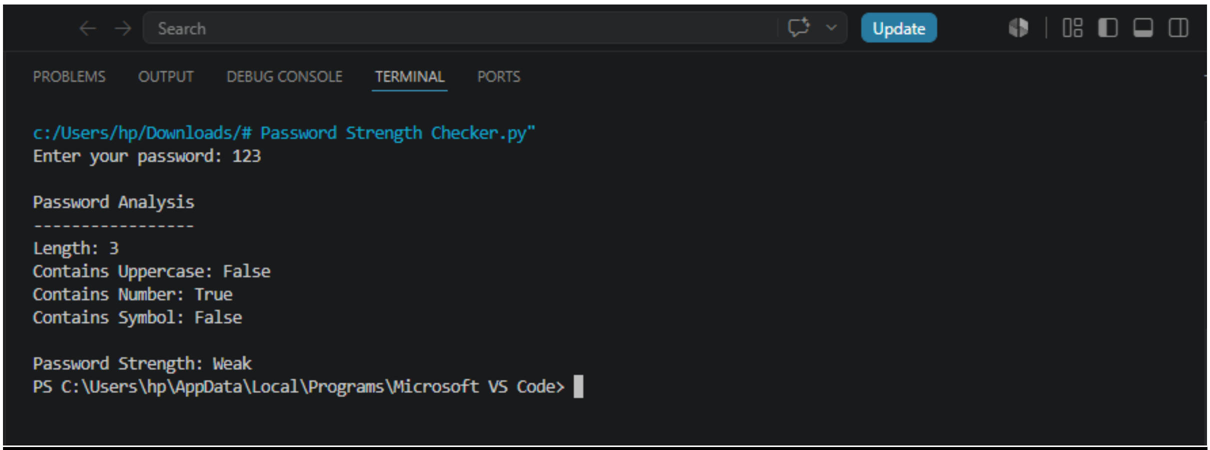
Password Evaluation Criteria

- Minimum length of 8 characters
- Contains uppercase letters
- Contains lowercase letters
- Contains numbers
- Contains special characters
- Checks against common passwords
- Provides feedback for improving weak passwords

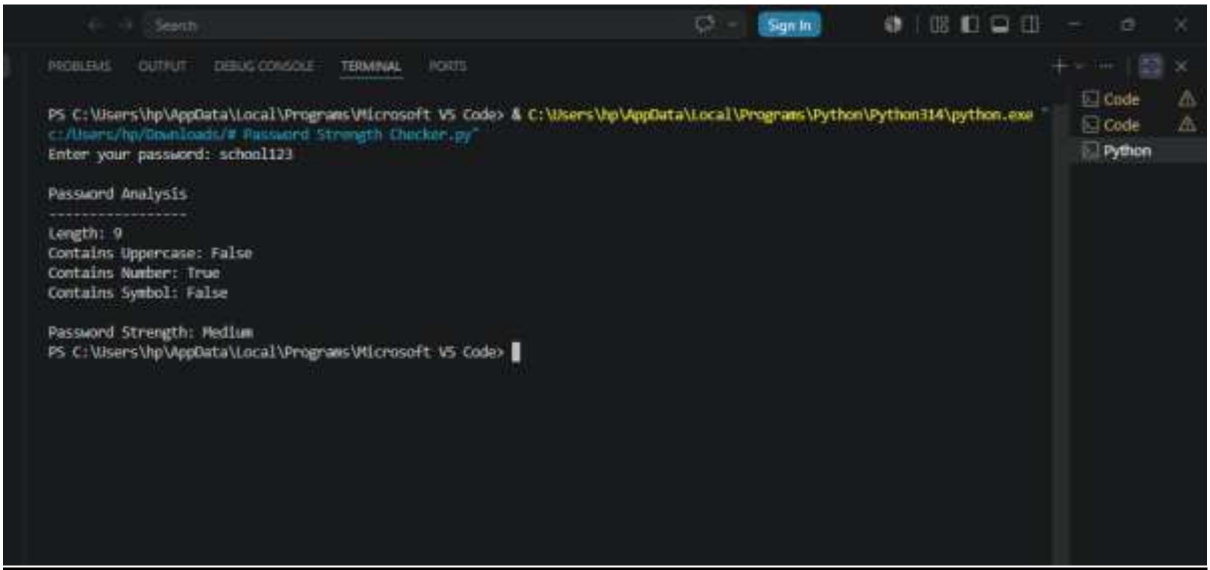
Example Outputs

Password	Strength	Reason
password	Very Weak	Common password
123456	Very Weak	Common password
Cyber2026	Medium	Missing special character
Cyber@2026	Strong	Meets all requirements

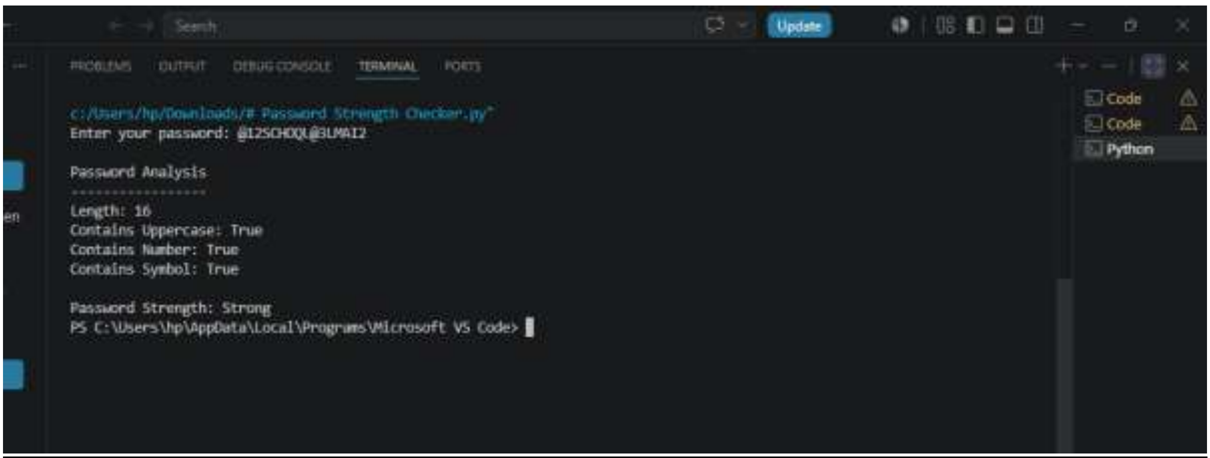
Weak Password:



Medium Password:



Strong Password:



3. Port Scanning Using Nmap

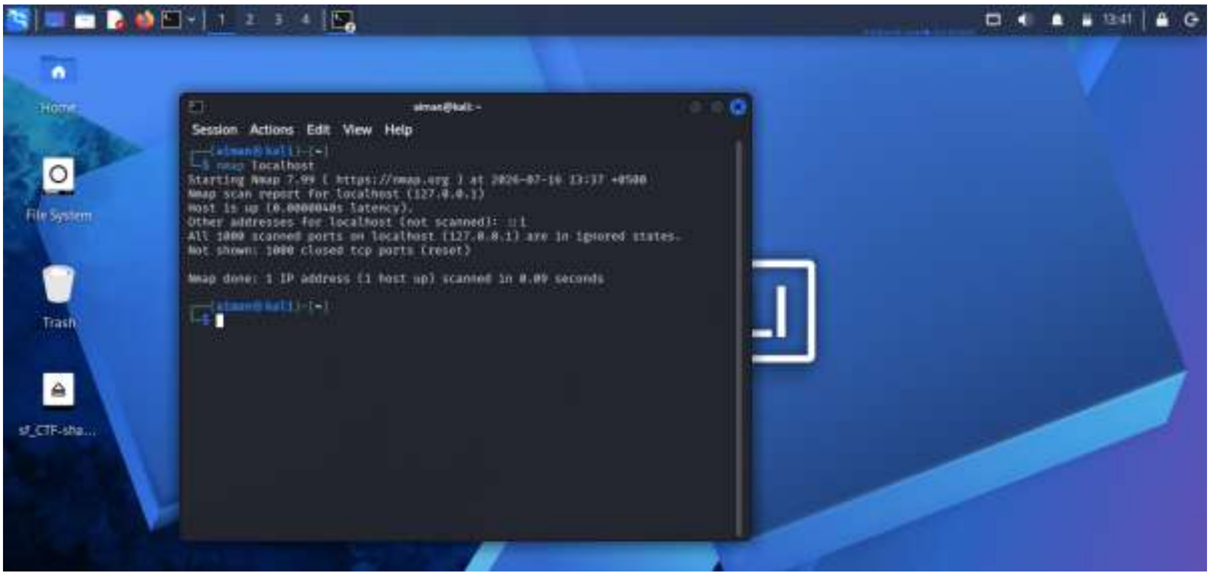
Tool Used

Nmap (Network Mapper) was used to scan localhost (127.0.0.1) for open TCP ports.

Command Executed

nmap localhost

Output



Scan Results

Port	Status	Service
None	Closed	No services detected

Analysis

The Nmap scan of localhost showed that all 1000 commonly scanned TCP ports were closed. This indicates that no services were actively listening for incoming network connections. Since no open ports were detected, the system presents a smaller attack surface, reducing the likelihood of network-based attacks.

4. Security Reflection

Weak passwords are one of the most common causes of unauthorized account access because they can be easily guessed or cracked using dictionary and brute-force attacks. Using long, unique passwords with a combination of uppercase letters, lowercase letters, numbers, and special characters greatly improves security.

Open ports allow network communication with running services. While some services require open ports to function, unnecessary or unsecured open ports can provide attackers with potential entry points into a system. Regularly reviewing open ports and disabling unused services helps reduce security risks.

This task demonstrated that both password security and proper network configuration are essential components of a secure system. Strong authentication combined with a minimal attack surface significantly improves overall cybersecurity.

5. Conclusion

This internship task provided practical experience with password strength evaluation and basic network scanning. It highlighted the importance of creating strong passwords and minimizing exposed network services. These are fundamental cybersecurity practices that help protect systems against unauthorized access and common attacks.